

# Problem

Nonprofit reporting is fragmented, late, and difficult to explain.

# Synthetic data model

8 fictional countries | 6 programs | 240 sites | 24 periods | 23,740+ fact records

# Architecture

Generator -> quality engine -> relational store -> FastAPI -> Streamlit -> exports

# Data quality

Completeness, validity, consistency, uniqueness, timeliness, referential integrity, freedom from errors

# Executive dashboard

Enrollment, outcomes, freshness, quality score, risks, latest period

# Funding lookup

Safe filters by synthetic funding source, program, site, and country

# Analytics / early warning

Rolling trends, IQR flags, missingness, bounded forecast, diagnostic indicators

# Migration / reconciliation

Matched, mismatched, rejected rows with readiness context

# Extension model

Planned / Extensible: Teams, Slack, WhatsApp, Telegram, Discord; not implemented

# Limitations / next steps

Synthetic only; add auth, governance, configurable rules, and PostgreSQL tests for